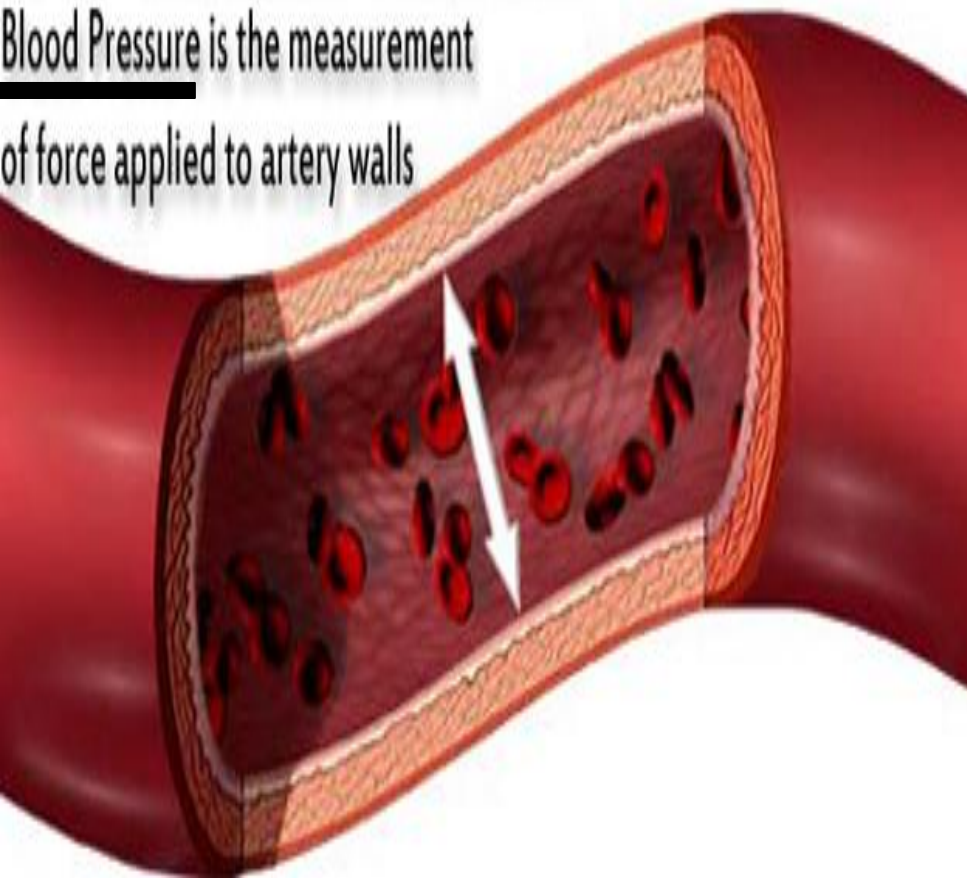
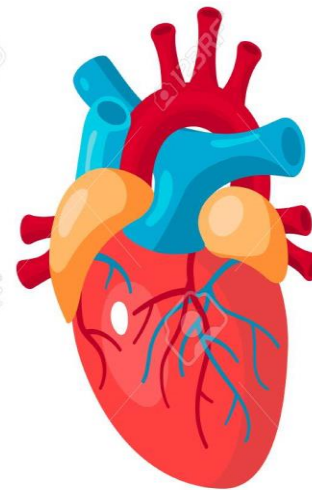


Blood Pressure is the measurement
of force applied to artery walls



Hypertension

HIGH BLOOD PRESSURE

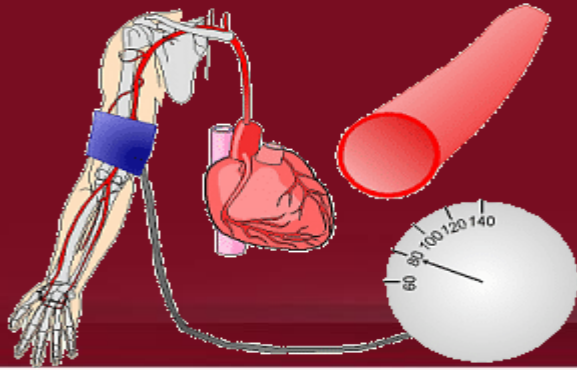


normal artery



narrowed arteries lead to
high blood pressure

$$BP = CO \times PR$$



Hypertension



-Hypertension is defined as sustained increase in BP.

-Hypertension occurs when **systolic BP > 130** mm Hg or **diastolic BP > 80** mm Hg on at least two occasions.

-Systolic: The top number, measures the pressure in the arteries when the heart beats (when the heart muscle contracts).

-Diastolic: The bottom number, measures the pressure in the arteries between heartbeats (when the heart muscle is resting between beats and refilling with blood).



Blood Pressure Category:

Category	Systolic	Diastolic
Normal	less than 120	less than 80
Pre-hypertension	<u>120</u> – 139	<u>80</u> – 89
(Hypertension) Stage 1	<u>140</u> – 159	<u>90</u> – 99
(Hypertension) Stage 2	<u>160</u> or higher	<u>100</u> or higher

Classification of BP, based on report of the seventh report of the Joint National Committee (JNC 7).

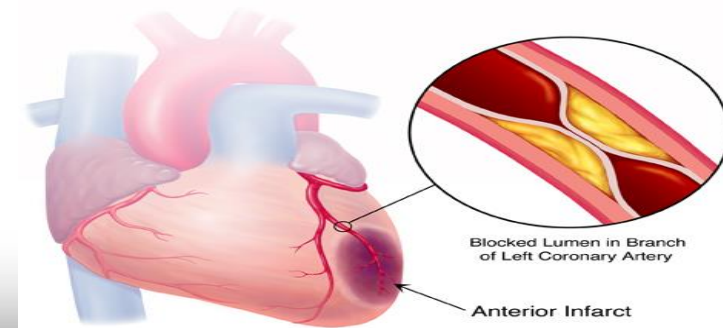
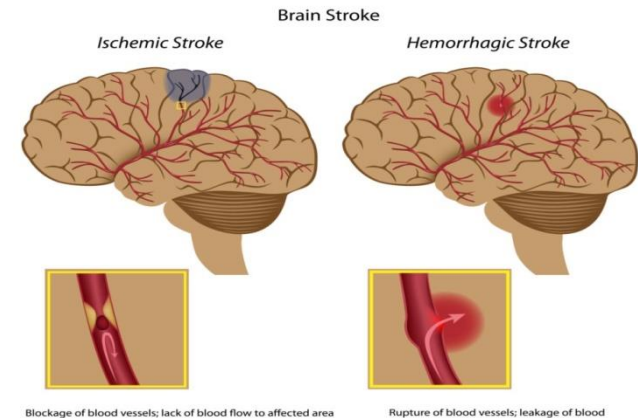
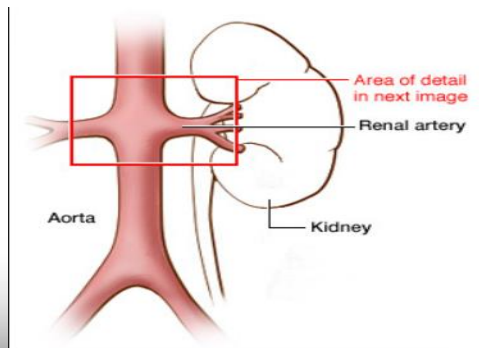
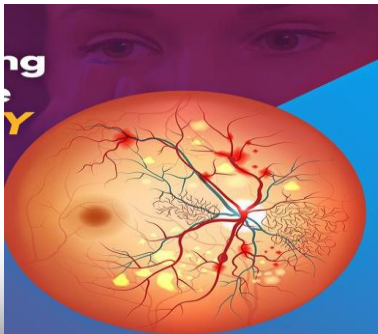
Why hypertension should be treated?



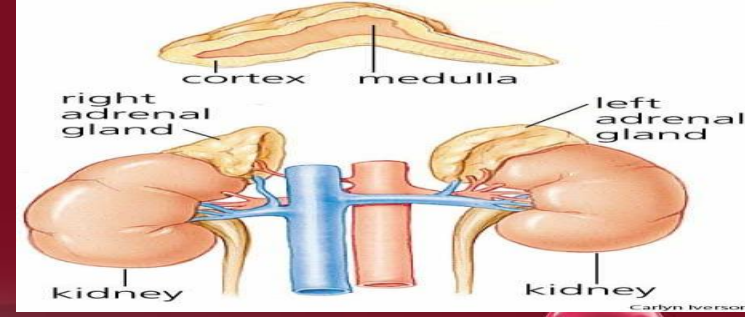
"**silent killer**" increases the rate of morbidity and mortality.

Chronic hypertension leads to:

- **Brain strokes**
(ischemic or hemorrhagic).
- Heart failure & Myocardial Infarction
- Renal failure & retinopathy .



Etiology of hypertension:



Primary (essential) hypertension	Secondary hypertension
About 80-90 % (Idiopathic)	About 10-20%
<u>Risk factors:</u> - ↑ salt intake, stress and smoking. - Overweight. - Black people & middle age male with high risk.	<u>Causes:</u> 1-Endocrine disease as: - Pheochromocytoma -Cushing disease - Hyperthyroidism. 2- drugs: Oral contraceptive 3- Renal disease.

Note: In case of secondary hypertension, you must treat the cause of hypertension (curative treatment).

Factors & parameters affecting blood pressure:

$$\bullet \text{BP} = \text{CO} \times \text{PR}$$

$$\bullet \text{CO} = \text{SV} \times \text{HR}$$



A-Stroke volume (SV): The amount of blood pumped by the left ventricle of the heart in one beat.

-Stroke volume depends on → **preload** and **afterload** and **contractility**

1-Preload: how much volume of the blood in the ventricle at the end of diastole (EDV).

2-Afterload: also known as the peripheral resistance (PR), is the amount of resistance the heart must overcome to open the aortic valve and push the blood volume out into the systemic circulation.

3- Contractility





$$\text{BP} = \text{CO} \times \text{PR}$$

-Cardiac output and peripheral resistance, in turn, are controlled mainly by two overlapping control mechanisms:

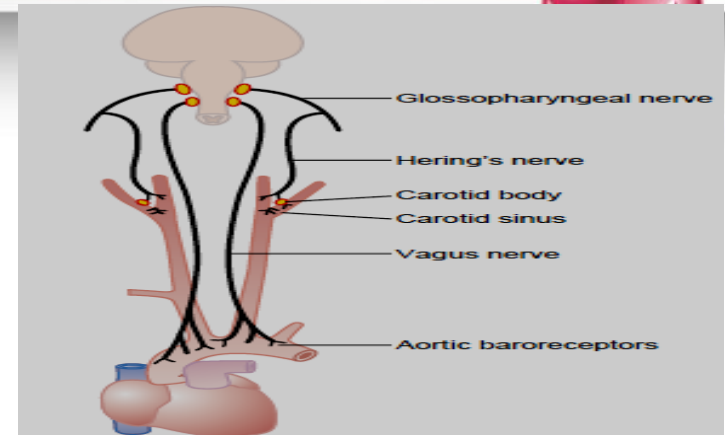
1-Baroreflexes→ Baroreceptors

2- Renin-angiotensin-aldosterone system

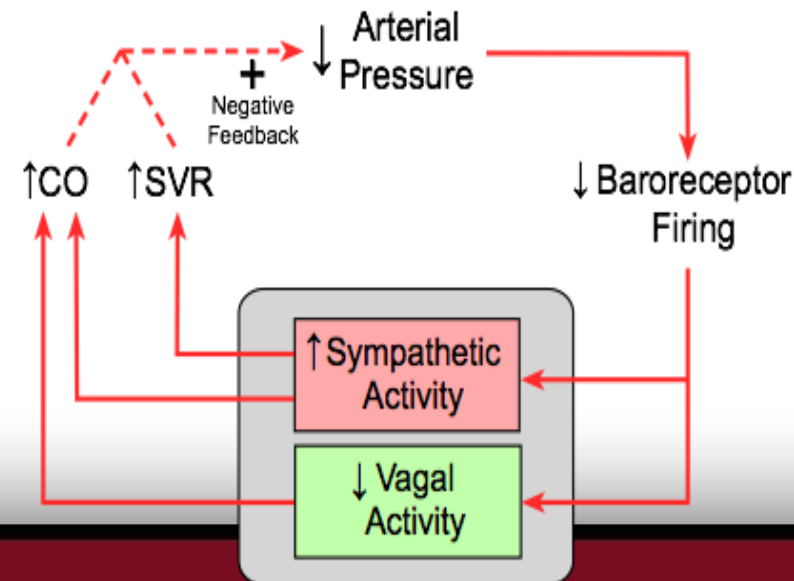
1-Baroreceptors and the sympathetic nervous system



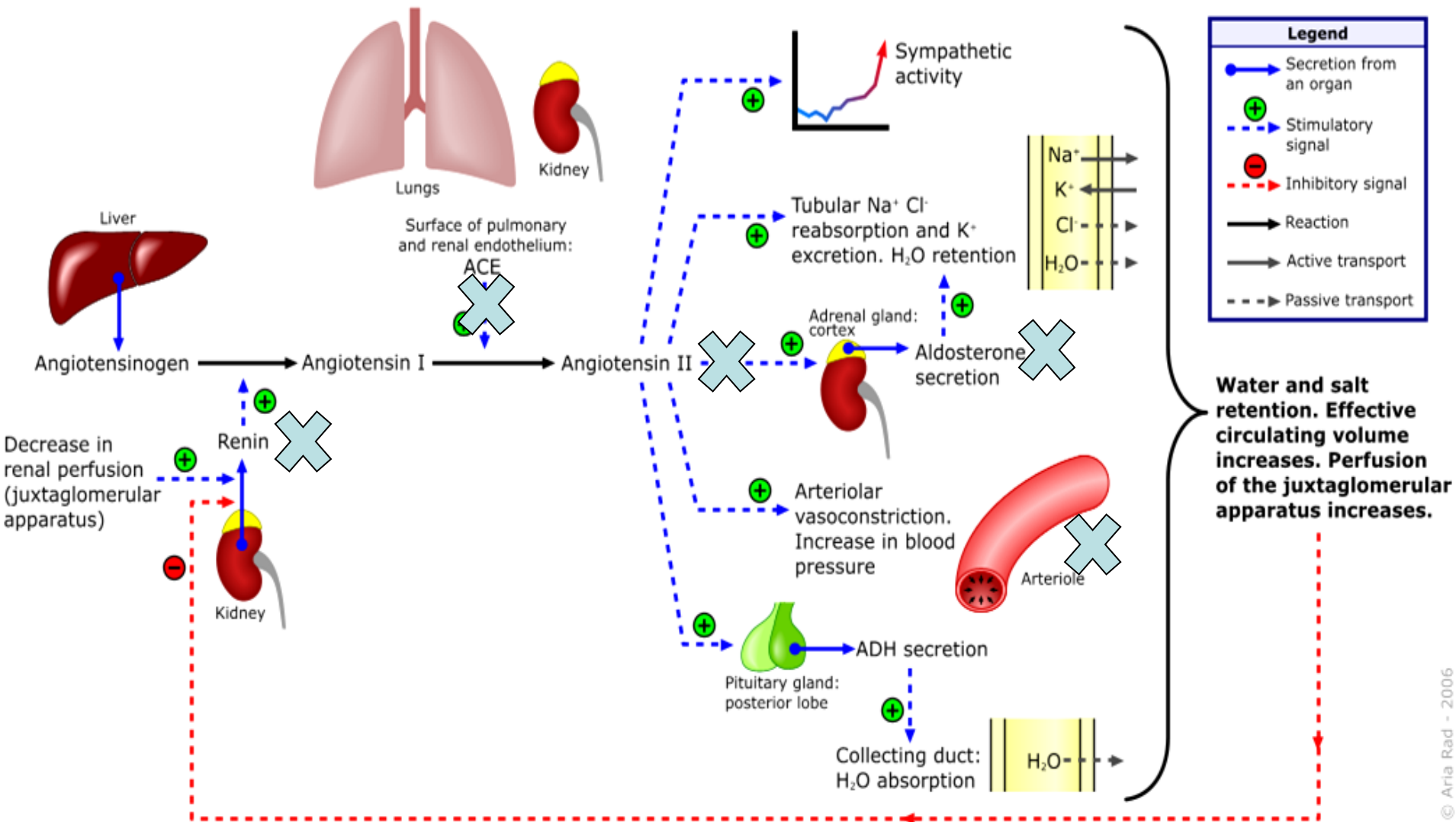
- $\uparrow\uparrow$ **BP** $\rightarrow$ distension of the aorta and the carotid arteries $\rightarrow$ stimulates baroreceptors $\rightarrow$ impulses vasomotor center $\rightarrow$ $\downarrow\downarrow$ **Sympathetic stimulation and $\uparrow\uparrow$ Parasympathetic stimulation**



- $\downarrow\downarrow$ **BP** $\rightarrow$ less distension or stretch of the aorta and the carotid arteries $\rightarrow$ $\downarrow\downarrow$ baroreceptor stimulation $\rightarrow$ $\downarrow\downarrow$ impulses vasomotor center $\rightarrow$ $\uparrow\uparrow$ **Sympathetic stimulation and $\downarrow\downarrow$ Parasympathetic stimulation**



2- Renin-angiotensin-aldosterone system (RAAS).



Other factors controlling blood pressure :



Vascular endothelium

- Nitric oxide** → vasodilator
- Prostacycline** (PGI₂) → vasodilator
- Bradykinine** → vasodilator
- Endothelin** → vasoconstrictor



Hormones

- Aldosterone and cortisol**
- Thyroxin**



Bosentan is a endothelin receptor antagonis

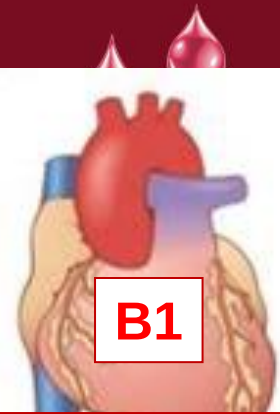


Antihypertensive drugs

α_2 agonist Centrally acting)
clonidine & alpha methyldopa.

A-Sympatholytic
drugs

$$BP = CO \times PR$$

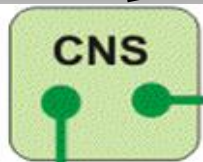


β blockers

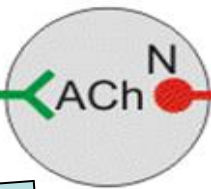
α blocker

α_1

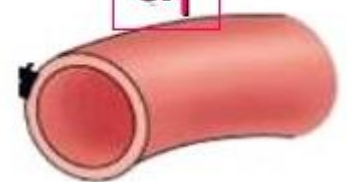
$\alpha + \beta$ blockers (labetalol).



Ganglion blockers
trimethaphan.



**Adrenergic neuron
blockers**
guanethidine, reserpine,
methyldopa.



1- Centrally acting drugs: **Clonidine & α - Methyl dopa**

i) Clonidine:



Mechanism:

-Presynaptic **α 2 agonist** $\rightarrow$ $\downarrow\downarrow$ release sympathetic outflow $\rightarrow$ $\downarrow\downarrow$ NE.

Uses:

-Hypertension complicated with renal disease $\rightarrow$ not $\downarrow$ RBF or GFR

NOTE: causes Na/H₂O retention so used in combination with diuretics

-Rebound hypertension occurs following abrupt withdrawal of clonidine.
The drug should, therefore, be withdrawn slowly.

ii) α -Methyl dopa :



Mechanism:

- It enters sympathetic nerve ending $\rightarrow$ converted into **α -methyl NE** which displace NE from synaptic vesicle & when exocytose less effect on α_1 receptors less V.C.
- Strong effect on α_2** $\rightarrow$ $\downarrow$ NE release.

Uses:

- Used in hypertension in pregnancy.
- Hypertension complicated with renal disease $\rightarrow$ not $\downarrow$ RBF or GFR.

Side effects of Clonidine & α -methyldopa:

- Sedation, Drowsiness and postural hypotension.
- Clonidine** $\rightarrow$ dry of nasal mucosa **α methyl-dopa** $\rightarrow$ hemolysis.

2) α -Blockers:

Prazosin & Terazosin & Doxazosin

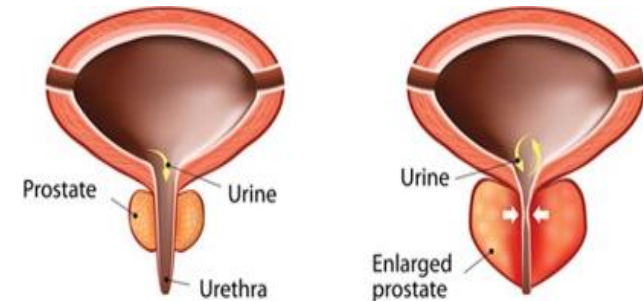


Mech : competitive $\alpha 1$ -blocker $\rightarrow$ V.D $\rightarrow$ $\downarrow$ PR $\rightarrow$ $\downarrow$ BP

Uses:

for mild or moderate hypertension.

-In combination with **proporanolol**



Side effects:

1-Orthostatic hypotension with reflex tachycardia and first-dose syncope
(so used in combination with propranolol)

2-Sexual and bladder dysfunctions

-**Tamsulosin:** is α_{1A} receptor used for patients with prostatic hyperplasia



3) β -Blockers:



- Non selective $\rightarrow$ **Propranolol** (block β_1 in heart and β_2 bronchi)
- Cardioselective $\rightarrow$ **Atenolol** & **Bisoprolol**
- Cardioselective with VD effect ($\uparrow \uparrow$ NO) $\rightarrow$ **Nebivolol**



Mechanism:

- Block β_1 in heart $\rightarrow$ $\downarrow$ cardiac output $\rightarrow$ $\downarrow$ B.P.
- Decrease release central sympathetic outflow.
- Decrease renin release from the kidney (Block β_1 in kidney) $\rightarrow$ $\downarrow$ angiotensin II & aldosterone.



Therapeutic uses:

- 1- Hypertension
- 2- Angina & myocardial infarction
- 3- Supraventricular arrhythmias
- 4- Migraine (**prophylaxis**)
- 5- Glaucoma (esp: **Timolol**)
- 6- Hyperthyroidism
- 7- **Anxiety**

N.B : β -blockers are more effective in white than black patients & in young than old patients. (**web search activity**)

Drug withdrawal: Abrupt withdrawal may induce angina, myocardial infarction.

Therefore, the dose of these drugs must be **tapered over 2 to 3 weeks** in patients with hypertension and ischemic heart disease.



Contraindications:

- Congestive heart failure (-ve inotropic)
- Bronchial asthma (if non-selective)
- Diabetes (esp.propranolol).

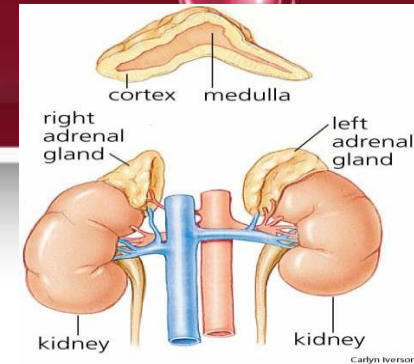
Note : Nebivolol can be used with chronic heart failure not for acute heart failure.

Adverse effects:

- CNS side effects (depression, fatigue hallucinations & night mares).
- Decrease HDL (risk for atherosclerosis).
- Bronchospasm .
- Impotence.

4) α & β -Blockers:

Labetalol & Carvedilol



Uses:

- Hypertensive crisis & pheochromocytoma.
- Carvedilol:** is effective as antihypertensive in heart failure.
- Labetalol:** can be used in pregnancy.

